When Protein Language Model Steering Appears to Work: Three Steering Studies and One Endpoint Analysis

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Abstract

Activation steering can change a protein language model's output without retraining the model, but a higher sequence score does not by itself establish control of a biological property. We retrospectively examine three completed ESM2-650M steering studies and one pre-steering endpoint analysis. In the endpoint analysis, a peptide-only composition score had held-out correlation $r = 0.427$ with a per-peptide label defined as the maximum normalized IC50 score observed across its assayed HLA-II alleles, but only $r = 0.100$ with measured T-cell response. Its full-length correlation changed from 0.379 under random folds to -0.323 under organism-grouped folds. No generation was run for this target. In the steering studies, an all-layer thermostability intervention retained 57 to 58 evaluable pairs in a shared low-strength range, but only 5 and 0 pairs at two higher strengths. An eSol soluble-fraction score contrast changed from 0.0125 to 0.00035 after dominant residues were excluded, and the resulting interval included zero. A disorder-score contrast was positive in three stored configurations, but a residue-exclusion interval excluded zero in only two. These cases motivate this reporting order, but they do not show that it generalizes beyond the four analyses or that steering improved any biological property.

1 Problem

1.1 Why apparent steering success needs staged checks

Protein language models learn statistical structure from large sequence collections and can support prediction and generation [Lin et al., 2023]. Activation steering modifies their internal representations at inference time. A direction is usually estimated from the difference between activations for high-label and low-label examples, then added to the residual stream during generation. Prior work applied this method to protein generation and optimization [Huang et al., 2025].

The intervention and its evaluation create distinct sources of error. A generated sequence may receive a higher score because it repeatedly inserts residues that occur directly in the scoring rule. A conditional score can also look favorable after low-complexity outputs have been removed from the comparison. Even before steering, a sequence score may perform well on one endpoint but poorly on the endpoint named in the claim. These problems concern the interpretation of the evidence, not only the strength of the intervention.

This paper asks which checks changed the conclusions of three completed ESM2-650M steering studies and one pre-steering endpoint analysis. We synthesize the evidence as four retrospective case studies rather than a catalog of attempted protein properties. The contribution is a proposed reporting sequence that separates four questions:

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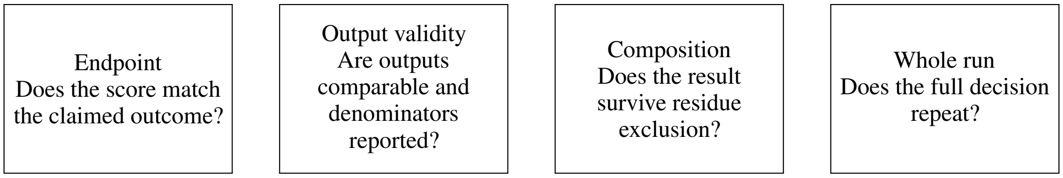

Figure 1: A staged reporting sequence derived from the completed case studies. A favorable downstream score does not override a failure at an earlier stage.

1. Does the score measure the endpoint named in the claim?
2. Do all arms produce comparable outputs, and are the denominators reported?
3. Does the result remain after dominant residue contributions are removed?
4. Does the full decision remain consistent across completed run configurations?

1.2 Evidence scope

The synthesis uses three completed steering studies and one completed pre-steering endpoint analysis. L52 compares a five-layer thermostability intervention with an all-layer intervention. L56 evaluates sequence-composition scores against several immune endpoints and stops before generation. L55 tests intrinsic-disorder steering across three stored whole-run configurations. L57 tests an eSol soluble-fraction score before and after excluding dominant residues. The three steering studies use ESM2-650M, matched-norm random directions, and single-shot masked filling.

The analyses are retrospective. Some checks were added after the original result had been inspected. We therefore use the studies to show how a check changes an interpretation, not to estimate the frequency of each failure mode in protein language model research.

2 Proposed Approach

2.1 A staged reporting sequence

Figure 1 shows the proposed order of the checks. Each stage limits the claim that can be carried into the next stage. A future evaluation should stop at the first unsupported inference. Later checks cannot repair an earlier failure.

The endpoint check compares the sequence score with the endpoint named in the claim. The output check records low-complexity output counts and the number of source proteins that remain jointly evaluable in every compared arm. The composition check removes the dominant substituted residues from the score and repeats the comparison. The whole-run check compares the final decision, including the composition check, across stored configurations.

2.2 Case-specific measurements

L52 evaluated 60 held-out source proteins at five intervention strengths. Both the five-layer and all-layer interventions used the same per-layer directions and the same source proteins. The saved analysis compared each intervention with a matched random direction using 10,000 paired bootstrap resamples. We inspect the jointly evaluable denominator before interpreting a conditional score.

L56 used records from the Immune Epitope Database [Vita et al., 2019]. Its source table contained 125,985 peptide–allele binding measurements, covering 15,362 unique peptides and 72 HLA-II alleles. The analysis grouped the measurements by peptide and used the maximum score observed across each peptide's assayed alleles. The normalized score was $1 - \log(\mathrm{IC}_{50}) / \log(50{,}000)$, so larger values corresponded to lower IC50 values. The fitted composition score used peptide sequence but not allele identity. Separate cohorts measured peptide presentation, T-cell response, and a full-length antigen label. For the full-length cohort, L56 also compared random-fold cross-validation with organism-grouped folds. No steering intervention was run because the endpoint check failed.

L55 used DisProt annotations [Quaglia et al., 2022] and the published TOP-IDP amino-acid scale [Campen et al., 2008]. L57 used eSol measurements [Niwa et al., 2009] and an absolute-charge

Table 1: Completed evidence used in the retrospective synthesis. The metrics differ because each check addresses a different inference.

Case	Apparent result	Failed or unstable check	Corrected interpretation
L56	A peptide-only score predicted an allele-aggregated assay label	Held-out r was 0.427 for each peptide's maximum normalized IC50 score and 0.100 for T-cell response; organism-grouped $r = -0.323$	Validate the endpoint named in the claim and group by source organism
L52	Five-layer steering appeared favorable at high strength	All-layer evaluable pairs fell from 57 to 58 in the shared range to 5 and 0	Compare methods only where both arms remain evaluable
L55 and L57	Learned directions increased their sequence scores	L55 residue exclusion passed in two of three stored runs; the L57 excluded interval crossed zero	Report composition and whole-run sensitivity with the main score

score motivated by a published recombinant-solubility model [Wilkinson and Harrison, 1991]. Each steering study compared the learned direction with a matched random direction over 150 held-out proteins. The saved decision used a 95% paired bootstrap interval over source proteins and then repeated the score after excluding the two most frequent substituted residues.

2.3 Interpretation rules

We report a conditional score only beside its evaluable denominator. An interval that includes zero is described as inconclusive, not as evidence of no effect. A drop under organism-grouped validation is described as consistent with confounding because grouping does not identify a unique cause. The three L55 artifacts are treated as completed whole-run configurations. Their files do not store seed metadata, and one legacy integer controlled several random processes. Differences among the runs therefore cannot be attributed to direction construction alone.

3 Observed Outcome

Table 1 summarizes the three failure classes and the corrected interpretation of each case.

3.1 Endpoint choice changed the L56 conclusion

The fitted peptide-composition score had held-out correlation $r = 0.427$ with a per-peptide label formed by taking the maximum normalized IC50 score across the alleles assayed for that peptide. The source table spanned 72 HLA-II alleles, but allele identity was not an input. The result therefore says that peptide composition was associated with this allele-aggregated dataset label. It does not estimate affinity for a named HLA-II allele or for a specified peptide–allele pair. The score's held-out correlation was $r = 0.100$ for measured T-cell response. Fixed motif scores showed the same broad loss of performance between the binding-assay and T-cell-response cohorts. Performance on the allele-aggregated assay label therefore did not justify a claim about immunogenicity.

The full-length cohort exposed a second problem. The composition model had $r = 0.379$ under random five-fold validation. When proteins from the same organism were kept in the same fold, the correlation was -0.323. The mean within-organism correlation was 0.056. This pattern is consistent with the model learning organism-associated composition rather than the intended endpoint. The evidence does not show that organism is the only cause, and it does not support a universal claim about sequence-based immune prediction. It was sufficient to stop this steering target before generation.

3.2 Low-complexity outputs made the L52 comparison unavailable

The original L52 decision selected an intervention strength from the full tested range. All 60 generated outputs retained numerical scores, but the historical low-complexity filter removed most all-layer outputs at high strength. At $\alpha = 1.0$, the all-layer arm retained only 5 jointly evaluable pairs. At

$\alpha = 2.0$, it retained none. The five-layer arm retained 58 and 57 pairs at those strengths. A favorable conditional score for the five-layer arm therefore did not establish that it was as effective as the all-layer intervention. The comparison was unavailable under the historical filter.

Restricting the interpretation to the shared low-strength range changed the result. At $\alpha = 0.5$, the five-layer contrast against its random control was 0.0236 with a 95% bootstrap interval of $[0.0202, 0.0270]$. The all-layer contrast was 0.0498 with interval $[0.0435, 0.0563]$. Across $\alpha = 0.1$ to 0.5, the five-layer intervention retained 43% to 59% of the all-layer effect. It produced a score change in this experiment, but the saved evidence does not support equivalence with all-layer steering.

3.3 Composition checks changed the L55 and L57 decisions

At $\alpha = 0.5$, the L55 learned-versus-random contrasts were 0.0497, 0.0373, and 0.0435 in the three stored whole-run configurations. The historical low-complexity threshold flagged 34, 47, and 33 of 150 learned-arm outputs, compared with 7, 8, and 13 of 150 random-control outputs. The reported contrasts used 116, 102, and 113 surviving pairs. The main TOP-IDP contrast was positive in all three configurations. After E and S were excluded, the corresponding contrasts were 0.0178, 0.0039, and 0.0181. The bootstrap interval included zero only in the middle configuration. The positive conditional contrast repeated, but the residue-exclusion decision did not repeat in every configuration.

L57 showed a stronger form of composition sensitivity. Its eSol soluble-fraction score contrast at $\alpha = 0.5$ was 0.0125 with interval $[0.0086, 0.0166]$. After excluding E and L, the two most frequent substitutions, the contrast was 0.00035 with interval $[-0.0028, 0.0035]$. This does not prove that the remaining effect is exactly zero. It shows that the original positive decision did not survive the stated composition check.

The saved one-run geometry result provides limited supporting context. The L55 and L57 directions had cosine similarity 0.376 overall and 0.556 to 0.666 in layers 30 to 32. This descriptive overlap has no control-vector distribution and does not establish a causal relationship between the two results.

4 Reason for Failure

4.1 Conditional scores hid denominator loss

A score computed only on surviving outputs answers a narrow question: how do the arms differ among source proteins for which both outputs remain evaluable? It does not describe the intervention over all attempted generations. In L52, the outputs remained scoreable, but a historical low-complexity rule left too few all-layer outputs for the high-strength comparison. Selecting strength by the conditional score allowed the five-layer method to look favorable where its comparison was unavailable. A steering report should place output counts, the evaluable denominator, and the conditional score beside each other, then limit method comparisons to a shared range.

4.2 The assay label did not match the claimed endpoint

The L56 target claim concerned measured T-cell response. The peptide-only composition score performed better against a per-peptide label defined as the maximum normalized IC50 score across the alleles assayed for that peptide. Because the score did not model allele identity, its performance describes an association with that allele-aggregated label. It is not generic binding affinity and does not estimate affinity for a specified peptide–allele pair. It also did not validate T-cell response. Random folds allowed organism-associated sequence composition to appear in both training and test sets. The resulting correlation may partly reflect source identity rather than the target endpoint. Endpoint definitions and grouping rules must therefore be fixed before a score is used to justify steering.

4.3 The scoring rule rewarded the dominant substitutions

Both TOP-IDP and the L57 eSol score are direct functions of amino-acid composition. Steering can raise such a score by repeatedly inserting residues that receive favorable weights. Residue exclusion is intentionally strict: it asks whether the decision survives after the largest direct contribution is removed. L55 showed that this decision can vary across whole runs even when the main score change remains positive. L57 showed that a favorable main interval can change to an interval that includes zero under the same check.

4.4 Practical reporting procedure

The cases suggest a concrete order for future evaluations. First, validate the scoring rule against the endpoint named in the claim and use grouped splits when source identity may leak. Second, report low-complexity output counts, evaluable denominators, and conditional scores as separate quantities. Third, test whether dominant residue substitutions account for the decision. Finally, repeat the complete decision process across run configurations. Repeating only the favorable main score is insufficient when an artifact check carries the conclusion.

5 Discussion and Limitations

The retrospective synthesis identified different limits because the checks protect different inferences. L52 concerns denominator loss under a low-complexity rule. L56 concerns whether a score represents the claimed endpoint. L55 and L57 concern direct dependence on amino-acid composition and sensitivity of the final decision. Together, they show why one successful-looking scalar is not a complete steering evaluation.

The evidence has important limits. Every steering result uses ESM2-650M, one masked-fill decoder, and one intervention family. The generation studies use a historical low-complexity threshold and conditional complete-case comparisons. Their saved files omit exact model revisions, source revisions, and some run metadata. The three L55 files do not record seed identity, so we treat them as legacy whole-run configurations. The L56 endpoint analyses use observational cohorts with different biological contexts and class structures. The grouped-validation change is consistent with confounding but does not identify a unique mechanism.

Several checks were selected or clarified after the corresponding outcomes were known. The bootstrap intervals describe stability over the fixed source proteins and should not be read as population-wide guarantees. The studies also contain multiple comparisons without a common familywise correction and use one matched random direction per saved run. No common audit was applied prospectively to every case, and no positive controls establish the specificity of the proposed reporting sequence. Most importantly, every reported endpoint is computational. No wet-lab assay establishes that a generated sequence has improved thermostability, intrinsic disorder, soluble fraction, immunogenicity, or another biological property.

6 Conclusion

The completed case studies motivate evaluating protein language model steering as a sequence of validity checks rather than by one favorable score. A method comparison can become unavailable when one arm loses most evaluable outputs. A sequence score may perform differently on the claimed endpoint or reflect source identity. A positive score can also depend on the residues that define the score and on the full run configuration. Across these four cases, reporting the checks together produced a narrower claim that matched the saved evidence. Whether the same order improves evaluations of other models or properties remains untested.

LLM usage disclosure

OpenAI Codex assisted with manuscript restructuring, prose editing, and source checks. Earlier project work used Claude (Anthropic) for code and manuscript drafting. The authors made the scientific decisions and checked the numerical claims against the saved artifacts.

Code and data availability

The study uses public protein and immunology resources together with saved model outputs. An anonymized artifact package will be provided if the submission system supports it. The review manuscript does not link to an identifying repository.

Ethics declaration

This computational study used public protein sequences and aggregated public annotations. It involved no new human participants, clinical intervention, or collection of personal data.

Author contributions

The authors designed the study, conducted the analyses, interpreted the results, and wrote the manuscript.

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Reproducibility statement

The manuscript reports completed artifacts already stored with the project. The evidence map identifies the source file for every numerical claim and records known provenance limits. No experiment rerun is part of the remaining manuscript work.

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